B.Tech. Sem-V

Fundamentals of Operating System Design (CEUC301)

UNIT 2: CONCEPTUAL QUESTION BANK

CPU Scheduling and Performance Evaluation

Unit / Weightage	Unit 2 | 4 hours
Question type	Conceptual, analytical and scenario-based

Syllabus coverage: Scheduling metrics and workload characterization; FCFS, SJF, Round Robin and Priority scheduling; multilevel and real-time scheduling (EDF and RMS); multilevel queue and feedback scheduling.

A. Scheduling Metrics and Workload Characterization

1. Why is CPU scheduling necessary in a multiprogramming operating system? Explain how it contributes to efficient resource utilization.
2. Define turnaround time, waiting time, response time, throughput and CPU utilization. Why does no single metric adequately describe scheduling performance?
3. Differentiate response time from turnaround time. Which metric is more important for an interactive application, and why?
4. Explain the relationship between waiting time, turnaround time and CPU burst time. Why can these measures lead to different conclusions about a scheduling policy?
5. What is workload characterization? Identify the workload properties that should be considered before selecting a CPU scheduling algorithm.
6. How do CPU-bound and I/O-bound processes differ, and why does their mixture influence scheduling performance?
7. Why should a scheduling algorithm be evaluated under multiple workloads rather than a single test case?
8. Explain the difference between fairness and high throughput. Can a scheduler maximize one while compromising the other?
9. How can scheduling overhead distort performance measurements, and what should be included in a fair comparison of algorithms?

B. FCFS, SJF, Round Robin and Priority Scheduling

10. Explain the operating principle of First-Come, First-Served (FCFS) scheduling. Under what workload conditions can it perform poorly?
11. What is the convoy effect in FCFS scheduling? Explain how a long CPU-bound process can affect shorter processes.
12. Why can Shortest Job First (SJF) minimize average waiting time when CPU burst lengths are known and the relevant assumptions hold?
13. What practical difficulty arises when implementing SJF in a general-purpose operating system where future CPU burst lengths are not known exactly?
14. Compare non-preemptive SJF with preemptive Shortest Remaining Time First (SRTF) in terms of responsiveness and preemption overhead.
15. Explain how Round Robin (RR) scheduling supports fairness and responsiveness in time-sharing systems.

16. What is the effect of choosing a very small time quantum in Round Robin scheduling? Discuss both benefits and costs.
17. What happens when the Round Robin time quantum becomes very large? Under what conditions does RR behave similarly to FCFS?
18. Explain priority scheduling and distinguish preemptive from non-preemptive priority scheduling.
19. What is starvation in priority scheduling? Explain how aging can reduce the risk.
20. Compare FCFS, SJF, RR and Priority scheduling with respect to fairness, response time, throughput and implementation considerations.
21. Why might an algorithm that produces a low average waiting time still be unsuitable for an interactive or deadline-sensitive system?

C. Multilevel and Real-Time Scheduling: EDF and RMS

22. What is multilevel queue scheduling? Explain why processes may be placed in separate queues based on their class or purpose.
23. How does multilevel queue scheduling differ from multilevel feedback queue scheduling in terms of process movement between queues?
24. Explain how a scheduler can allocate CPU time among queues using fixed priorities or time slicing.
25. What is real-time scheduling? Distinguish hard real-time requirements from soft real-time requirements.
26. Why is meeting a deadline different from merely minimizing average waiting or turnaround time?
27. Explain Earliest Deadline First (EDF) scheduling and the principle used to select the next task.
28. Explain Rate Monotonic Scheduling (RMS). Why are shorter-period periodic tasks assigned higher fixed priorities under RMS?
29. Compare EDF and RMS in terms of priority assignment, adaptability and assumptions about task behaviour.
30. Why do schedulability analysis and worst-case execution-time estimates matter in real-time systems?
31. Discuss why a scheduler's average-case performance may be insufficient evidence that a hard real-time workload is schedulable.
32. Identify situations in which EDF may be preferable to RMS, and explain what system assumptions must be considered before making that choice.

D. Multilevel Feedback Queue Scheduling

33. Explain the main design idea behind a multilevel feedback queue (MLFQ) scheduler.
34. How can MLFQ use observed CPU-burst behaviour to distinguish interactive tasks from CPU-intensive tasks?
35. Why are short time quanta often used in higher-priority queues and longer quanta in lower-priority queues?
36. Explain how promotion, demotion and periodic priority boosts can influence responsiveness and starvation.
37. What problems can occur if a CPU-intensive process is repeatedly demoted and receives very little CPU time?
38. How can a process attempt to exploit an MLFQ policy, and what design measures can reduce such behaviour?
39. Why does MLFQ require policy choices such as queue count, time quantum, demotion rules and boost interval?
40. Compare MLFQ with fixed multilevel queue scheduling in terms of flexibility, complexity and adaptation to workload changes.

E. Integrative and Higher-Order Conceptual Questions

41. A desktop system must keep interactive applications responsive while running long background computations. Recommend a scheduling approach and justify the trade-offs.
42. A batch-processing system prioritizes high throughput, while a user-facing system prioritizes response time. Explain why the same scheduling policy may not be optimal for both.

43. A system uses Round Robin but experiences excessive context-switching overhead. Explain which scheduler parameter could be adjusted and what performance trade-off should be evaluated.
44. A process with a long CPU burst delays many short jobs under FCFS. Compare at least two alternative policies and discuss their limitations.
45. A real-time controller misses deadlines despite reporting high CPU utilization. Explain why utilization alone does not establish deadline feasibility and identify the analyses required.
46. An organization must schedule periodic real-time tasks with known periods and execution-time estimates. Explain how it could evaluate RMS and EDF before deployment.
47. Design a conceptual experiment to compare FCFS, SJF, RR and Priority scheduling. Specify workload classes, metrics, and controls needed for a fair evaluation.
48. Critically evaluate the statement: 'The scheduling algorithm with the lowest average waiting time is always the best algorithm.' Use interactive, batch and real-time workloads in your argument.